X402 Forbidden Protocol Architecture Design

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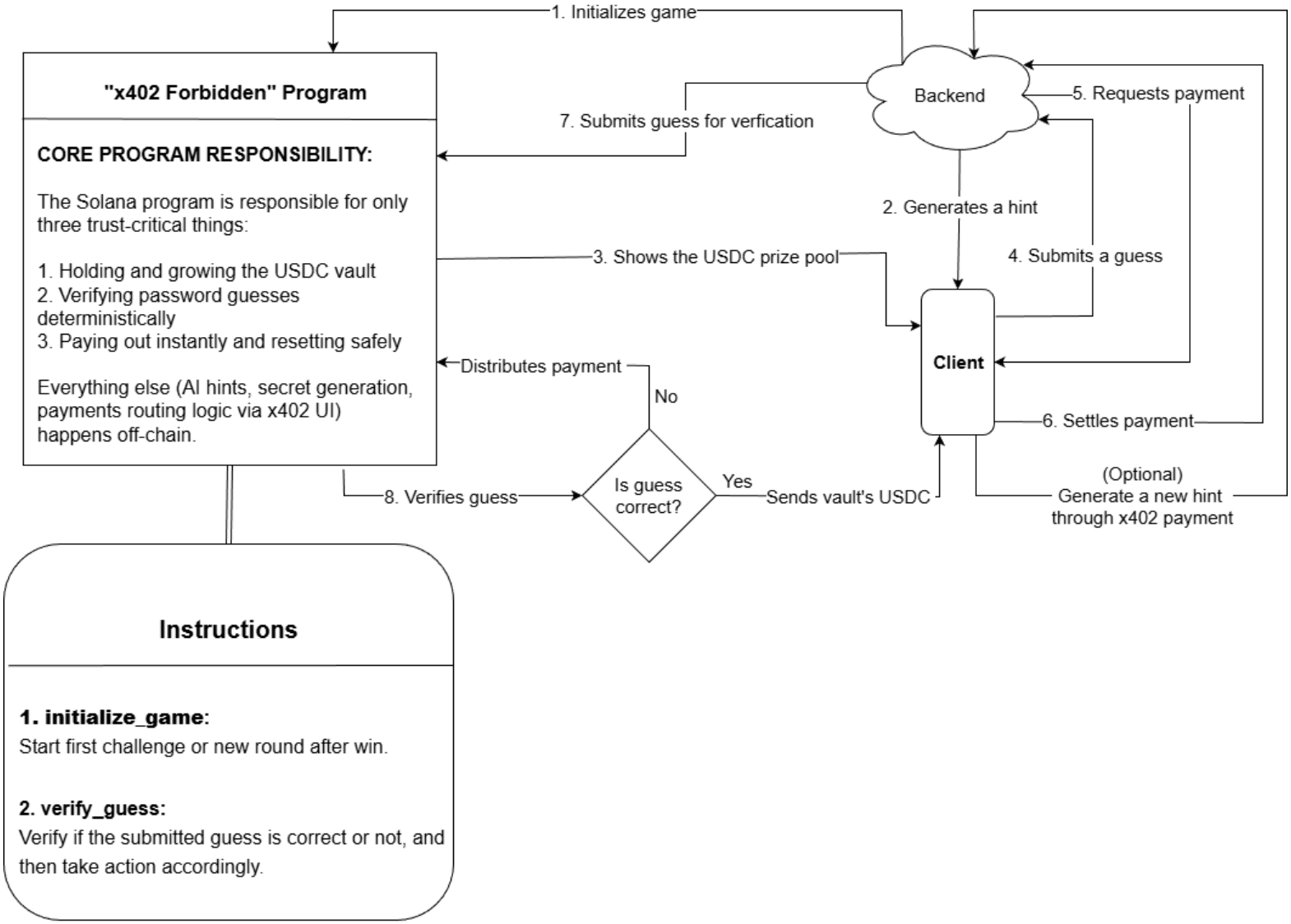
Program Interaction Matrix – Overview



Instructions

1. initialize_game:
Start first challenge or new round after win.

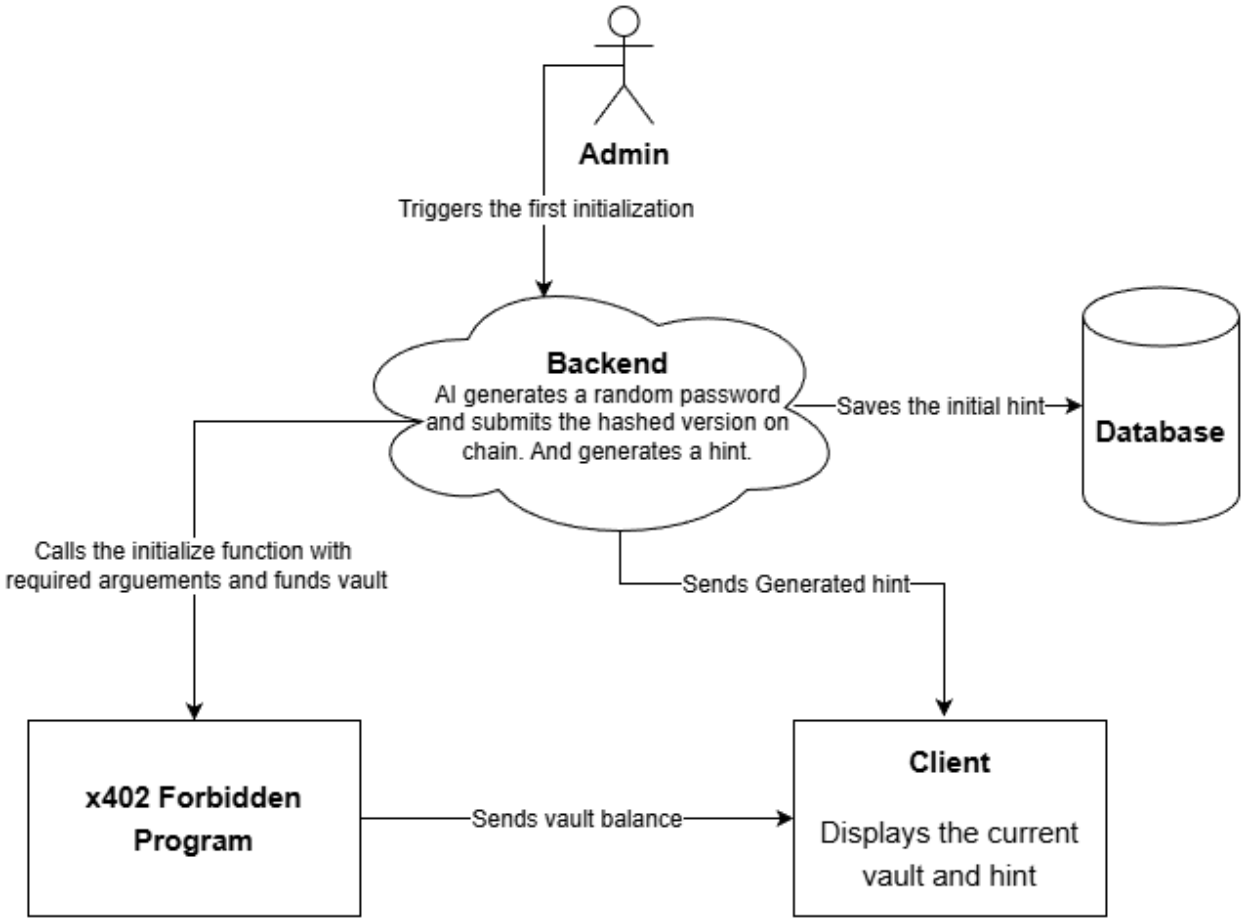
2. verify_guess:
Verify if the submitted guess is correct or not, and then take action accordingly.

Steps in brief:

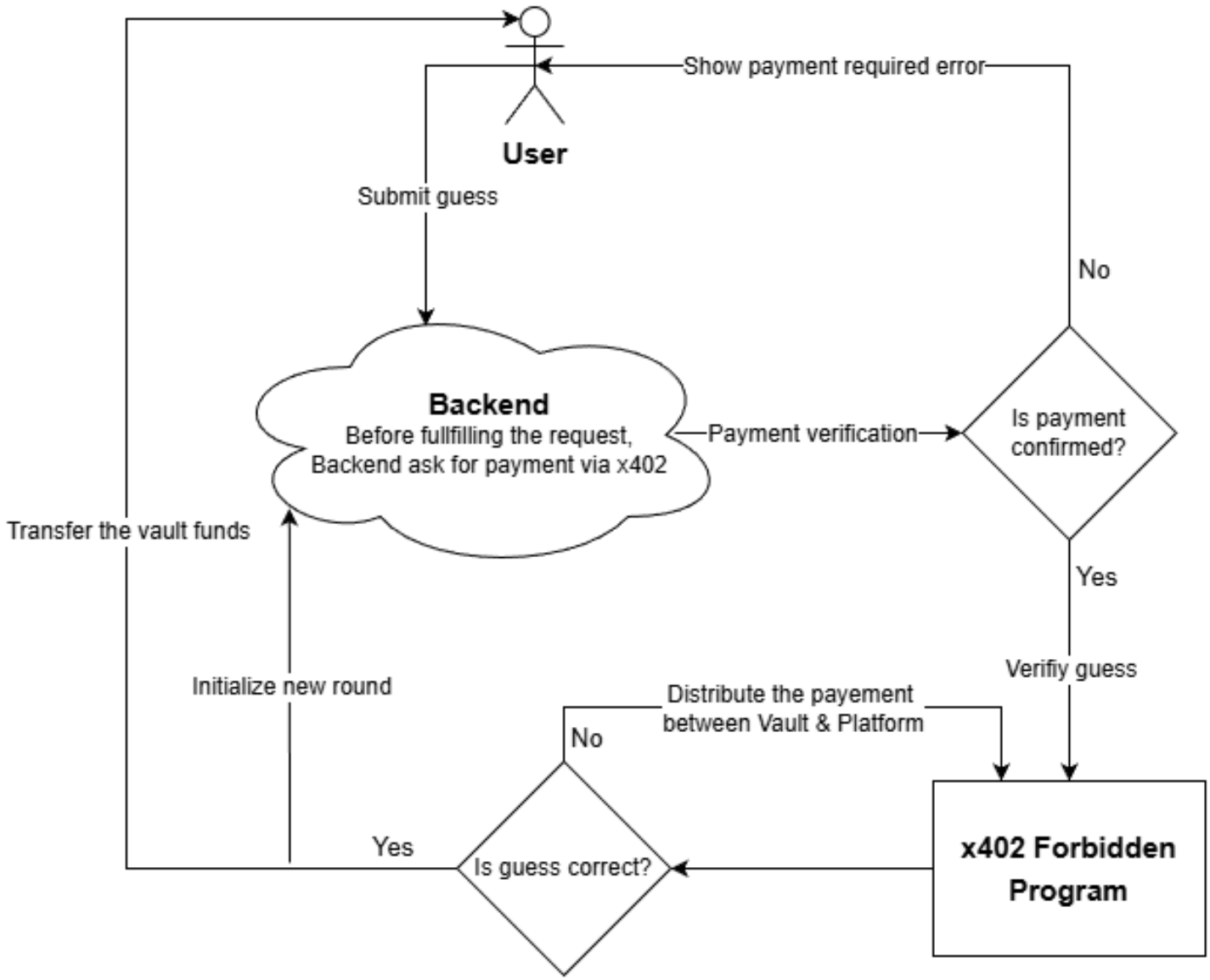
1. **Initialize game**: Backend generates and hashes the password & stores it on-chain along with some configurations. And funds the vault.
2. **Generates a hint:** Backend then generates the hint for the password.
3. **Shows the USDC prize pool:** The browser displays the current vault pool after getting it from program.
4. **Submits a guess:** The user takes a guess for the password and submits it to the backend.
5. **Requests payment:** The backend asks for a micro-payment for submission via x402 protocol.
6. **Settles payment:** The user settles the payment using their wallet.
7. **Submits guess for verification:** Once payment is confirmed, the backend submits the guess to program for on-chain verification.
8. **Verifies guess:**
   - **If guess is correct,** program transfers the prize pool to the user's wallet and the game is re-initialized.
   - **If guess is incorrect,** the payment is distributed among the prize pool and platform fees.

User Interaction Flow & External Integrations

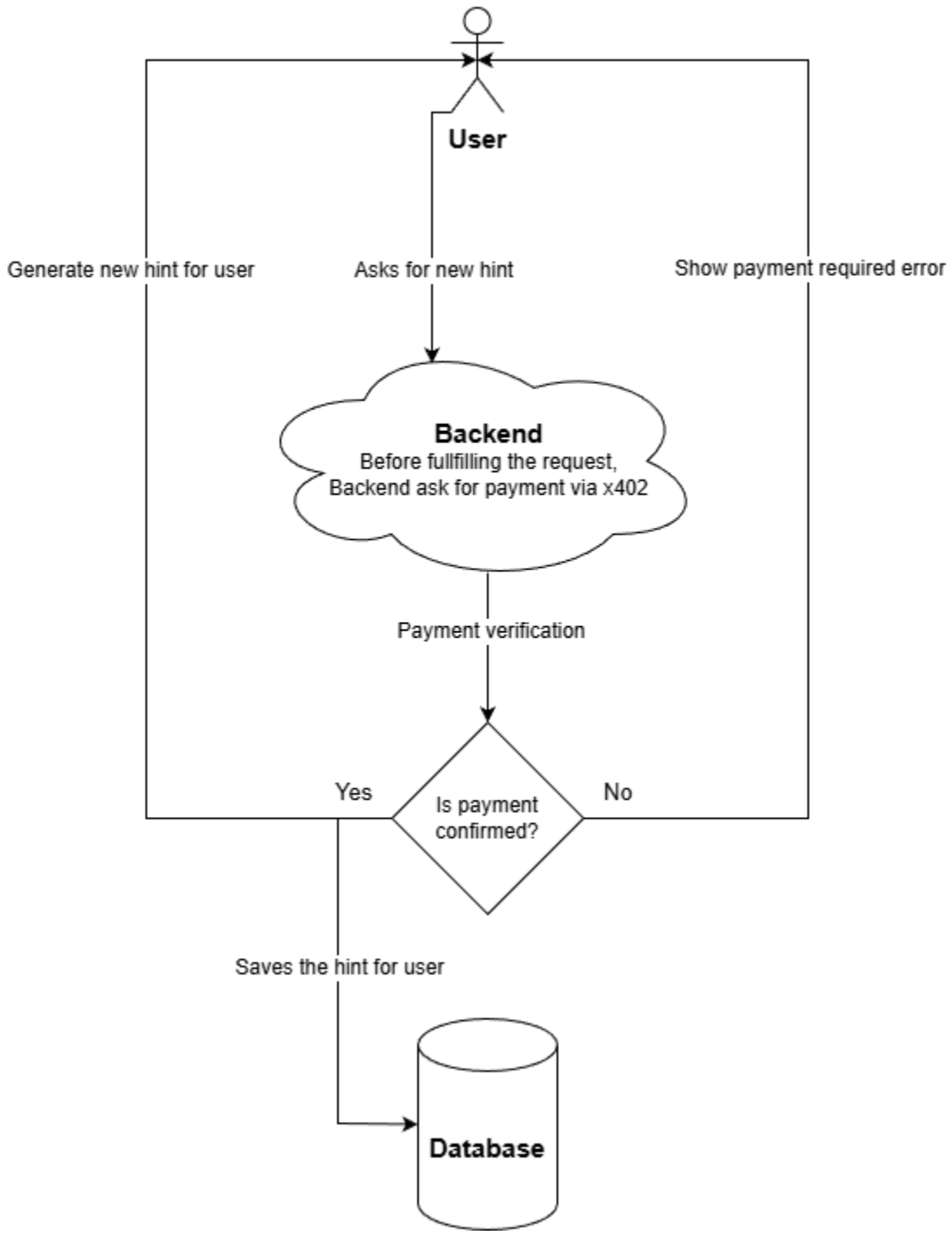
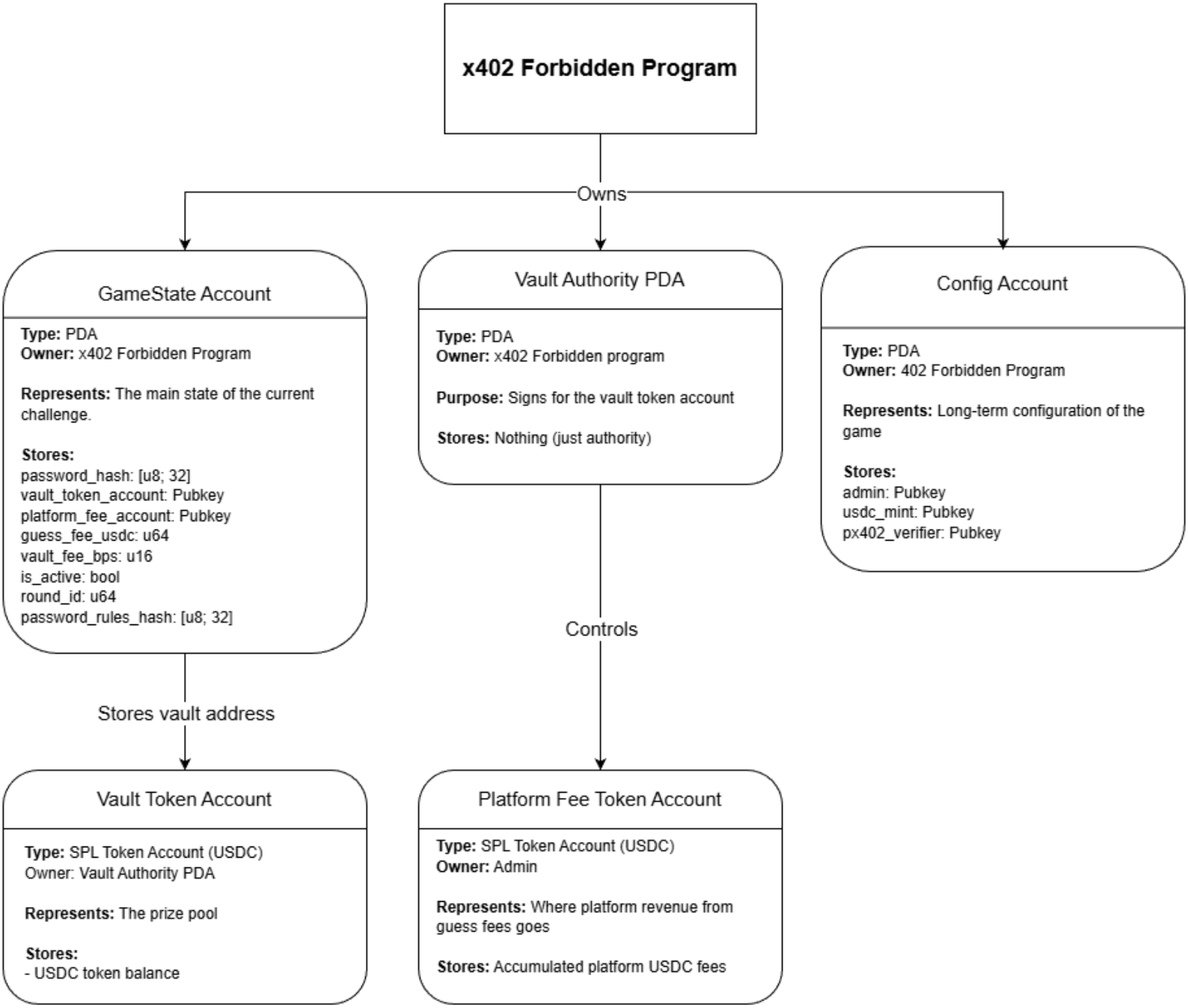
1. Initializing the game (First time)



2. Submitting guess and game conclusion



2. New hint generation



Account Management